

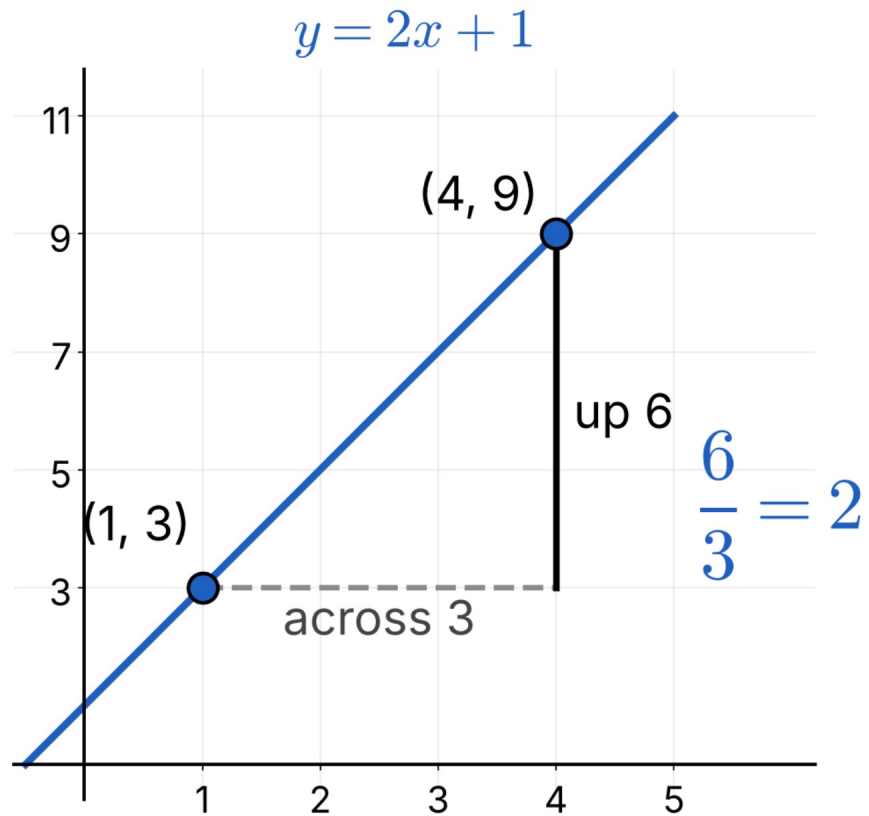
MATH FOR AI • EPISODE 0.5

Slope

Rise over run, and the seed of all of calculus.

Dr. Joe

Last time's challenge



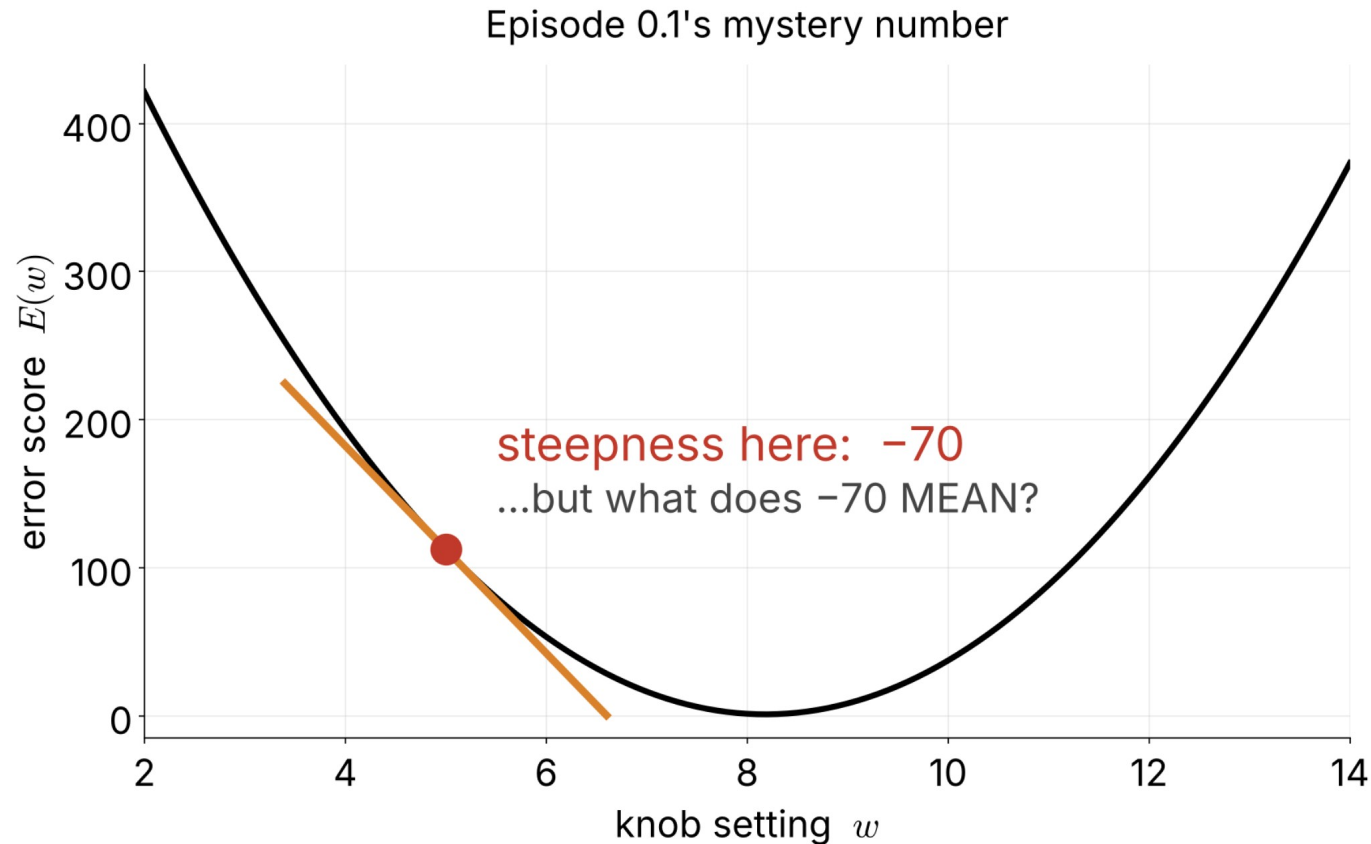
FROM (1, 3) TO (4, 9)

Up **6**. Across **3**. Divide:

$$\frac{9 - 3}{4 - 1} = \frac{6}{3} = 2$$

Where have you seen 2? It's the **w** in $y = \mathbf{2}x + 1$.

Episode 0.1's mystery number



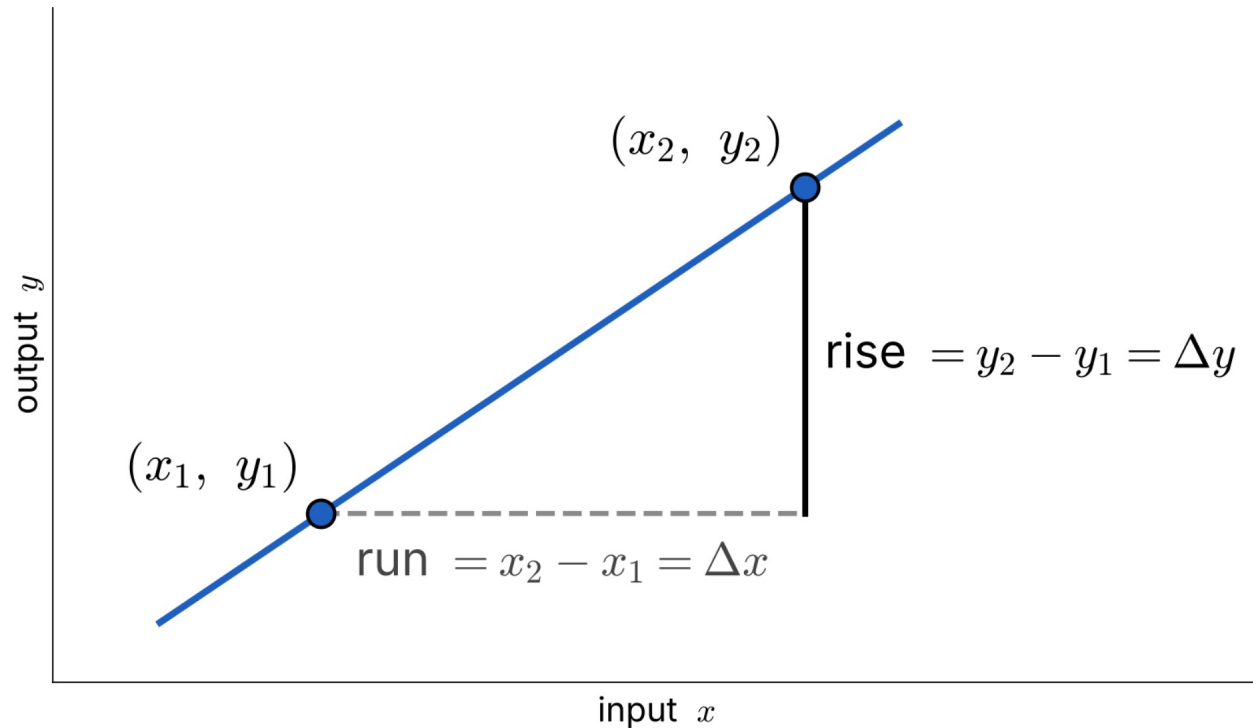
IN EPISODE 0.1

At $w = 5$, the valley's steepness was **-70**.

We used it. We never explained it.

Today: what it means, and how to check it with subtraction and division.

Rise over run



$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

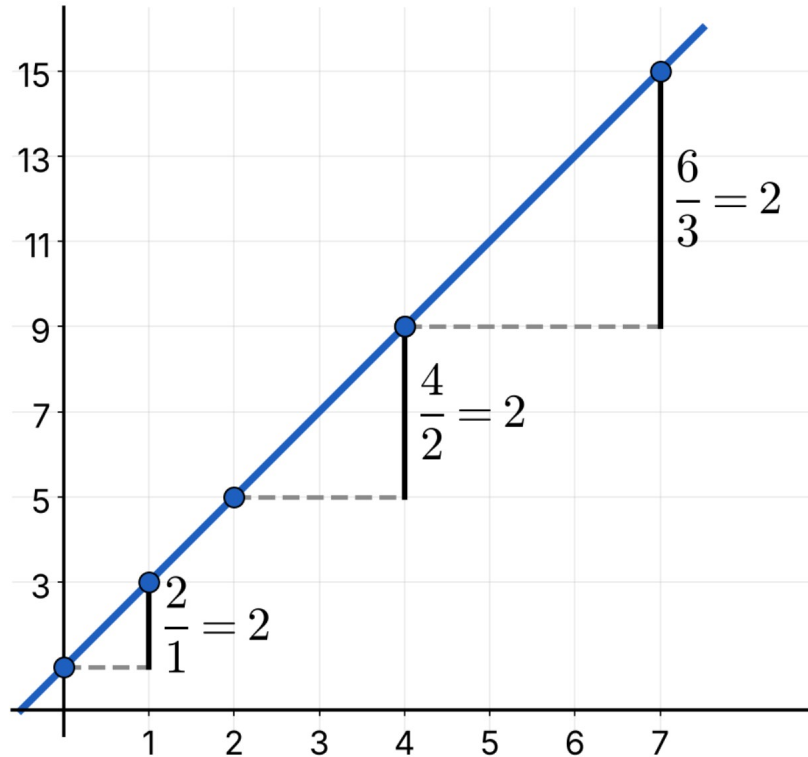
$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

Δ (capital delta) means "change in."

Slope = how much the output changes for each step of the input.

A line has one slope

$y = 2x + 1$: every staircase gives the same slope



ANY TWO POINTS

Small staircase or big one, the answer is **2**.

EITHER ORDER

$$\frac{3 - 9}{1 - 4} = \frac{-6}{-3} = 2$$

Just use the **same order** on top and bottom.

Why the slope of $y = wx + b$ is w

$$\frac{(w x_2 + b) - (w x_1 + b)}{x_2 - x_1}$$

rise over run, for any two points on the line

$$= \frac{w x_2 - w x_1}{x_2 - x_1}$$

the two **b**'s cancel

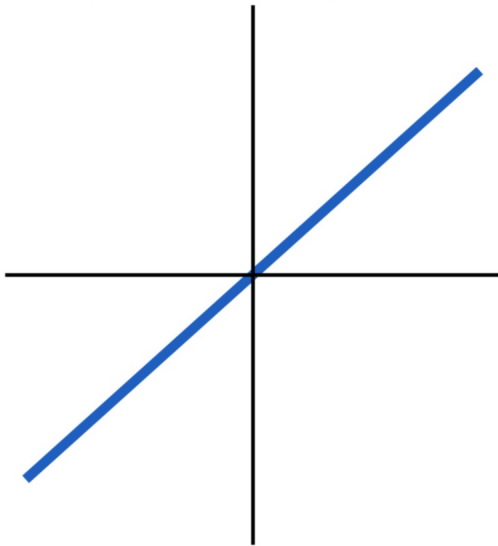
$$= \frac{w (x_2 - x_1)}{x_2 - x_1} = w$$

pull out **w**, then top and bottom cancel

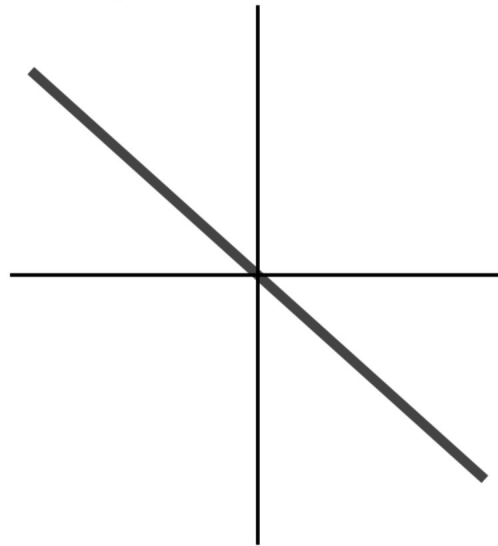
The tilt knob w IS the slope. Every time.

Four kinds of slope

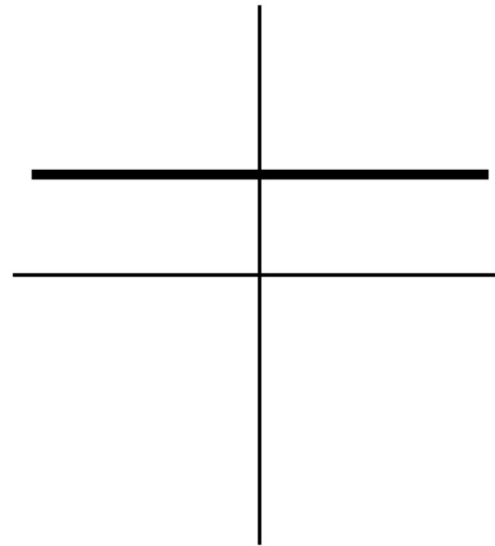
positive: uphill



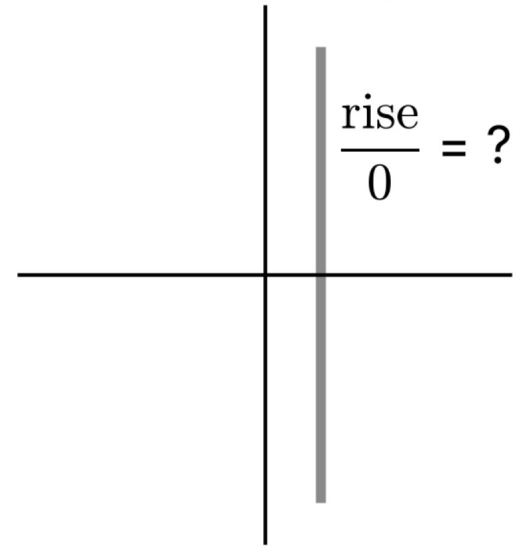
negative: downhill



zero: flat



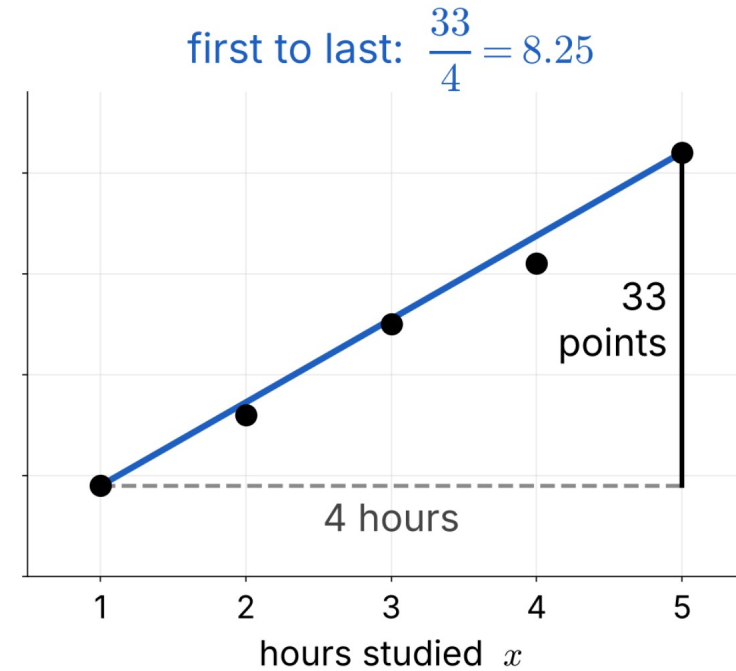
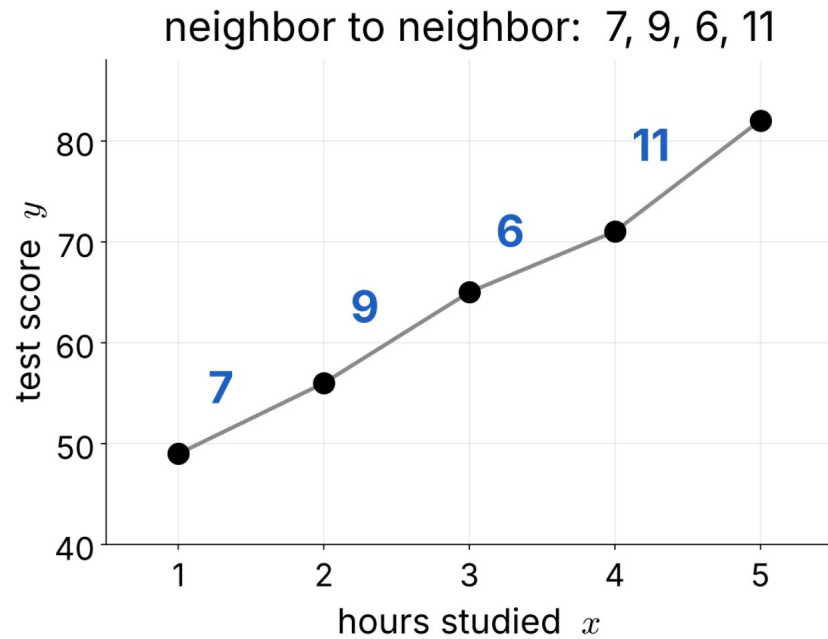
undefined: straight up



$$\frac{5 - 1}{2 - 2} = \frac{4}{0} = ?$$

Straight up: the run is **0**, and you can't divide by 0. No slope number at all.

Slope is a rate



$$\frac{82 - 49}{5 - 1} = \frac{33 \text{ points}}{4 \text{ hours}} = 8.25 \frac{\text{points}}{\text{hour}}$$

Units: output units **per** input unit.

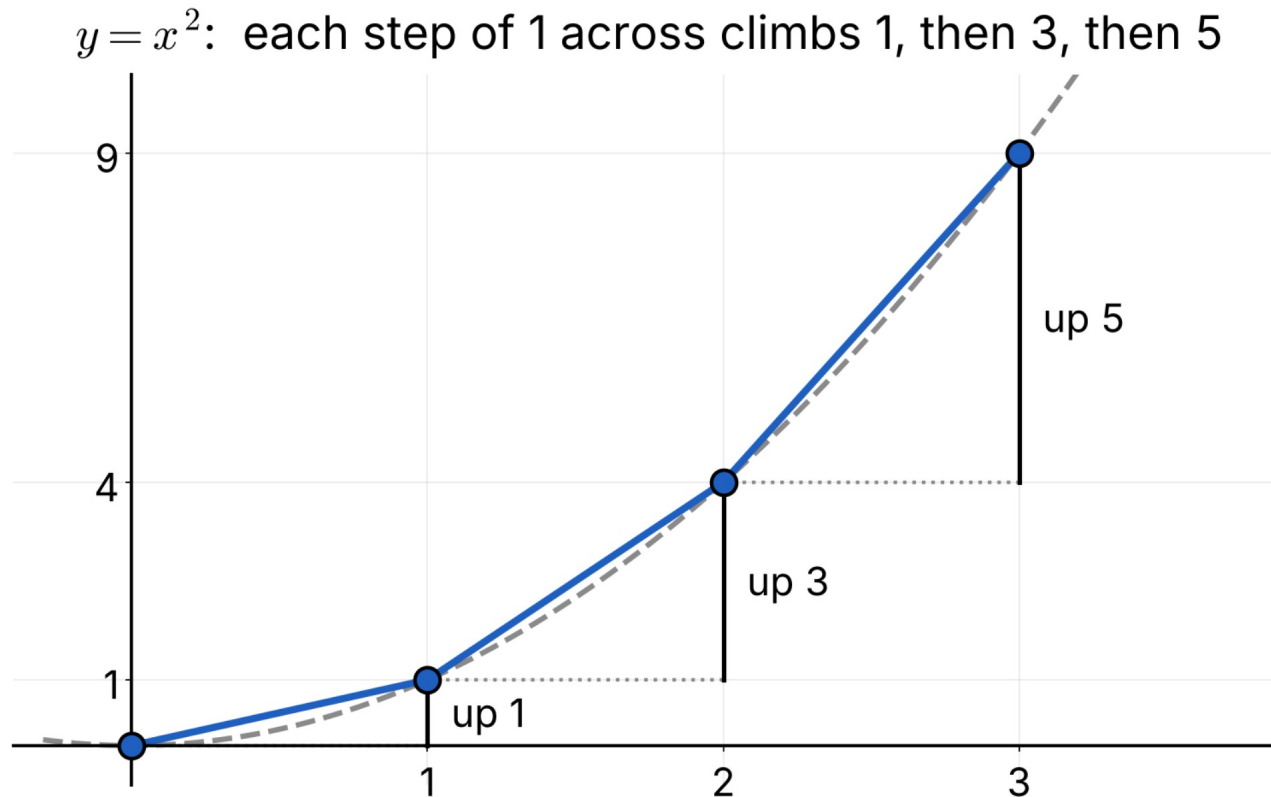
Slope in Python

```
def slope(p1, p2):
    x1, y1 = p1          # unpack the first point
    x2, y2 = p2          # unpack the second point
    rise = y2 - y1       # how far UP
    run = x2 - x1        # how far ACROSS
    if run == 0:
        return None      # run is 0: can't divide by 0
    return rise / run

print(slope((1, 3), (4, 9)))    # 2.0
print(slope((4, 9), (1, 3)))    # 2.0 (order doesn't matter)
```

Two points in, one number out. Rise over run is just a machine.

A curve's slope keeps changing



$$y = x^2$$

$$\frac{4 - 1}{2 - 1} = 3$$

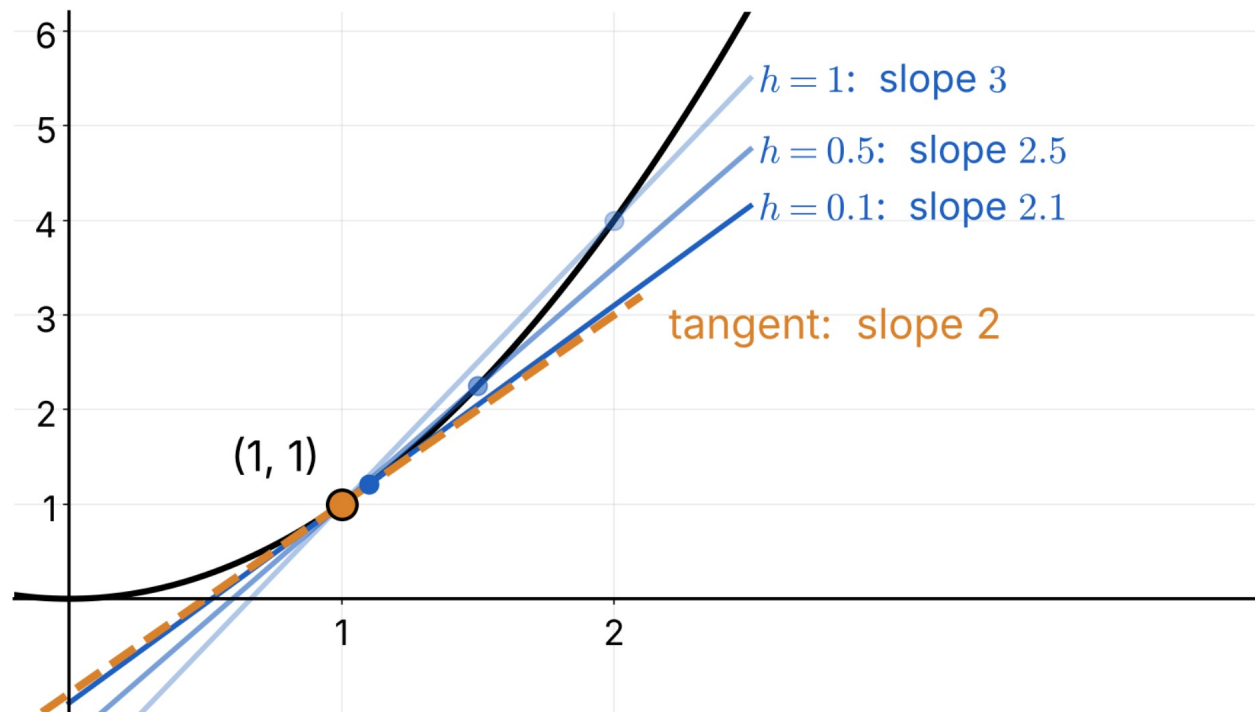
$$\frac{9 - 4}{3 - 2} = 5$$

Steeper and steeper. **No single slope.**

So: what's the slope **at one spot?**

Shrink the run

Shrink the run h : the line through two points turns into the tangent



$$\text{slope near } 1 \approx \frac{(1+h)^2 - 1^2}{h}$$

run h	slope near $x = 1$
1	3
0.5	2.5
0.1	2.1
0.01	2.01
0.001	2.001

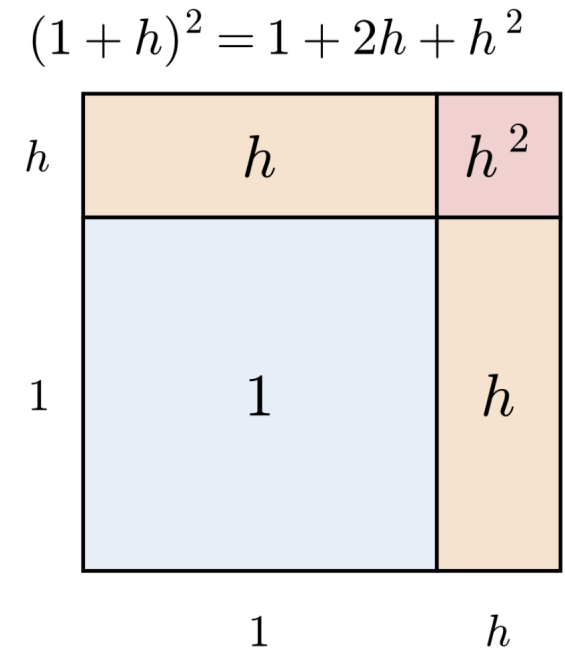
Settling on **2**.

Why it settles on 2

$$\begin{aligned}\frac{(1+h)^2 - 1^2}{h} &= \frac{1 + 2h + h^2 - 1}{h} \\ &= \frac{2h + h^2}{h} = 2 + h\end{aligned}$$

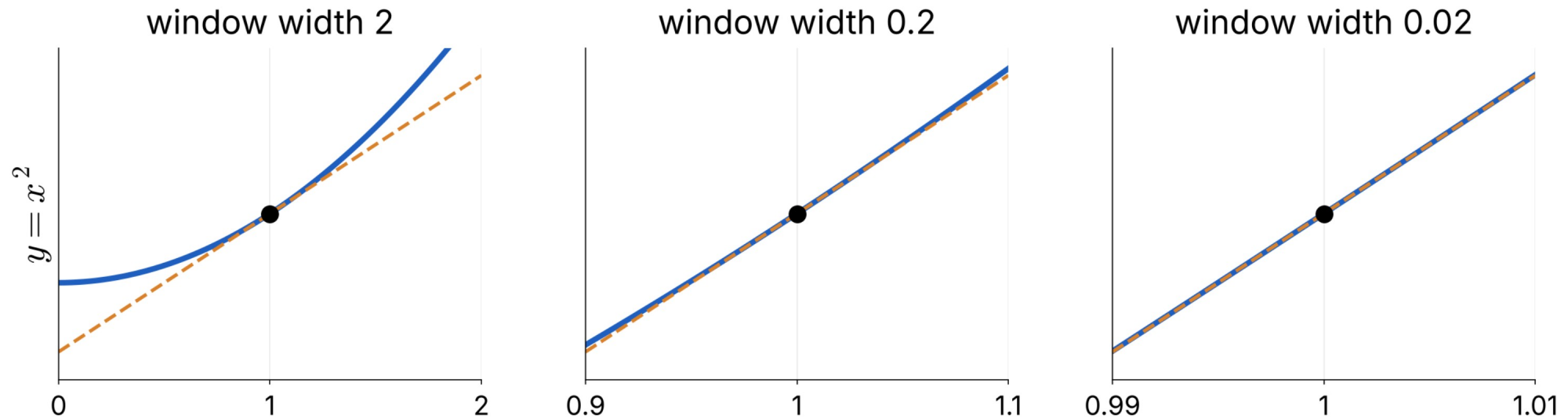
as $h \rightarrow 0$, $2 + h \rightarrow 2$

This is the seed of calculus. In Season 3 it gets its name: the derivative.



Zoom in, and a curve looks straight

Zoom in on (1, 1): the curve straightens into its tangent line



Up close, a smooth curve (no sharp corners) is almost a line, and its slope is the slope of that line.

The mystery, solved

THE ERROR VALLEY, NEAR $w = 5$

$$E(w) = 11w^2 - 180w + 737.4$$

$$\frac{E(5 + h) - E(5)}{h}$$

$$= -70 + 11h \rightarrow -70 \text{ as } h \rightarrow 0$$

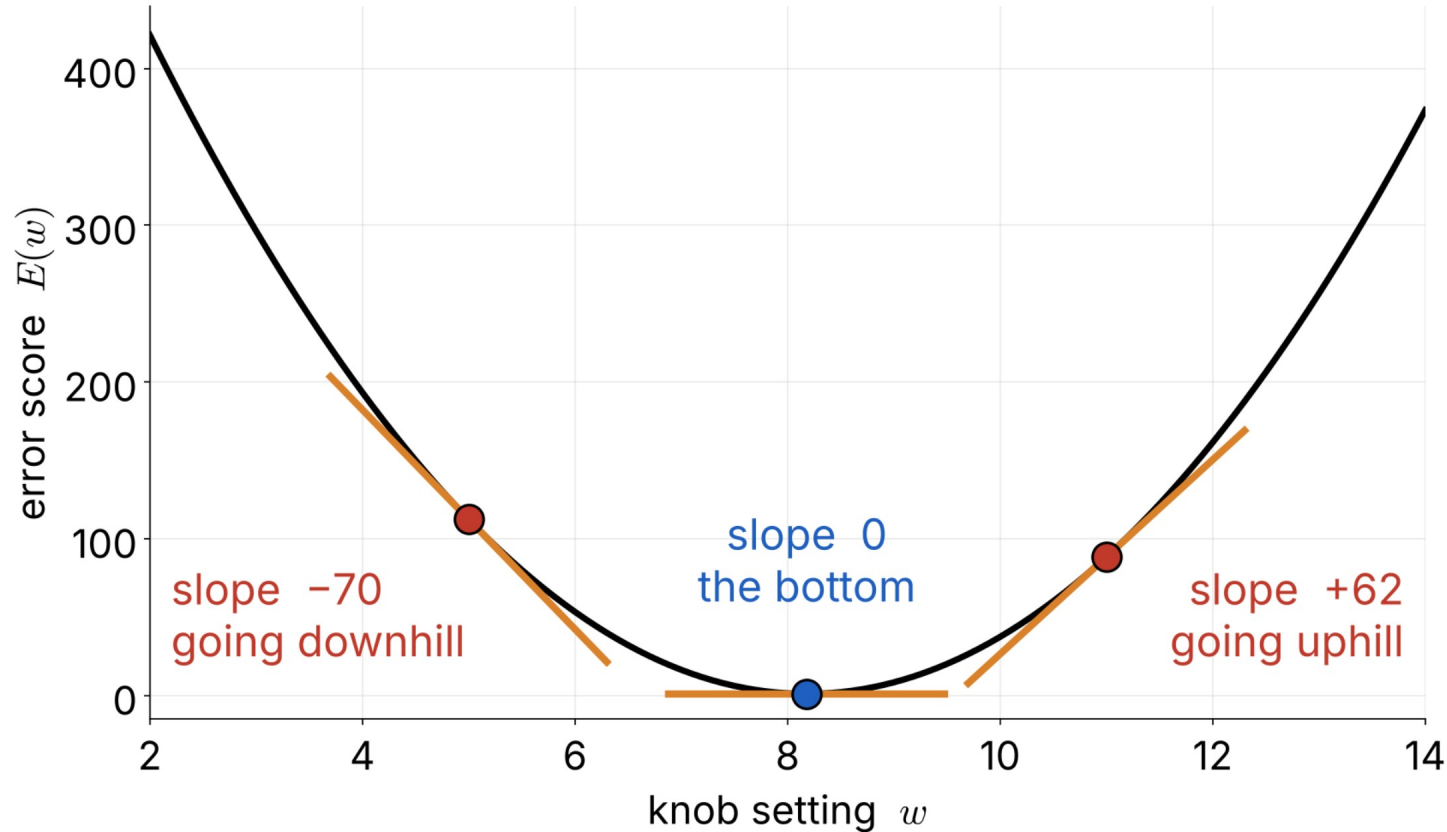
Same move as on x^2 : rise over a shrinking run.

Careful: two graphs. w is the slope of the prediction line. **-70** is the slope of the error valley.

run h	slope near w = 5
1	-59
0.1	-68.9
0.01	-69.89
0.001	-69.989

-70 is the slope of the valley at $w = 5$.

The sign is a compass

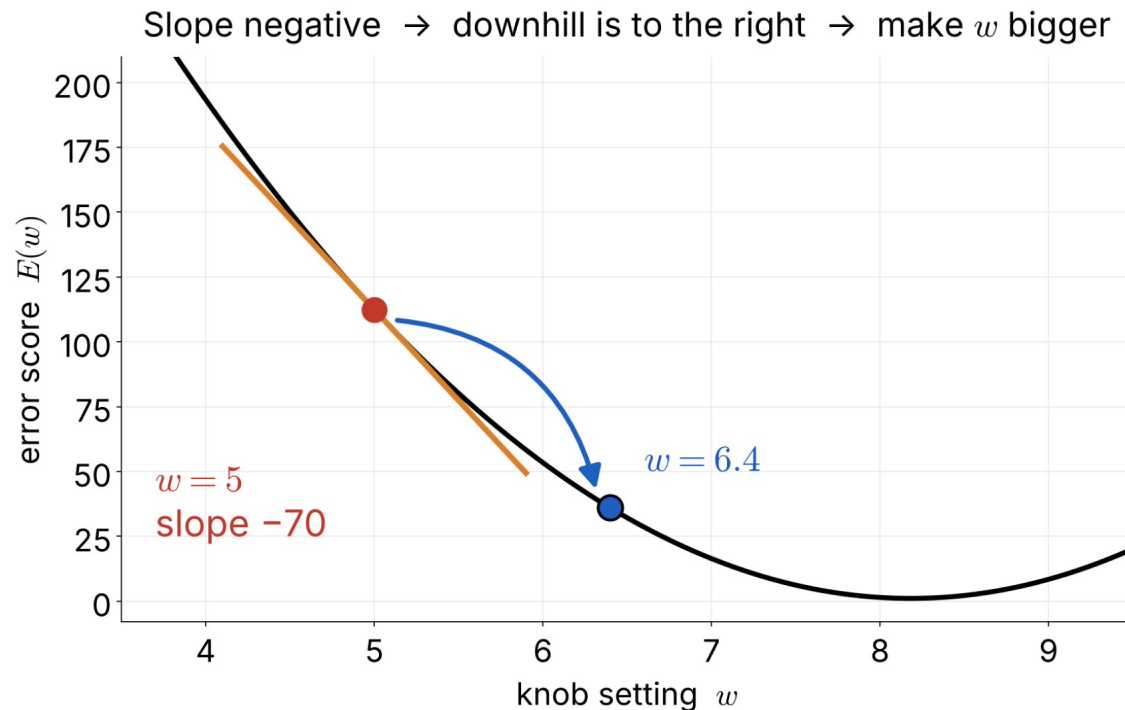


Negative: downhill is to the **right**.
Make w bigger.

Positive: downhill is to the **left**.
Make w smaller.

Zero: flat. In this U-shaped valley,
that's the bottom.

One step downhill, explained



EPISODE 0.1'S UPDATE RULE

$$w_{\text{new}} = w - \eta \cdot (\text{valley slope})$$

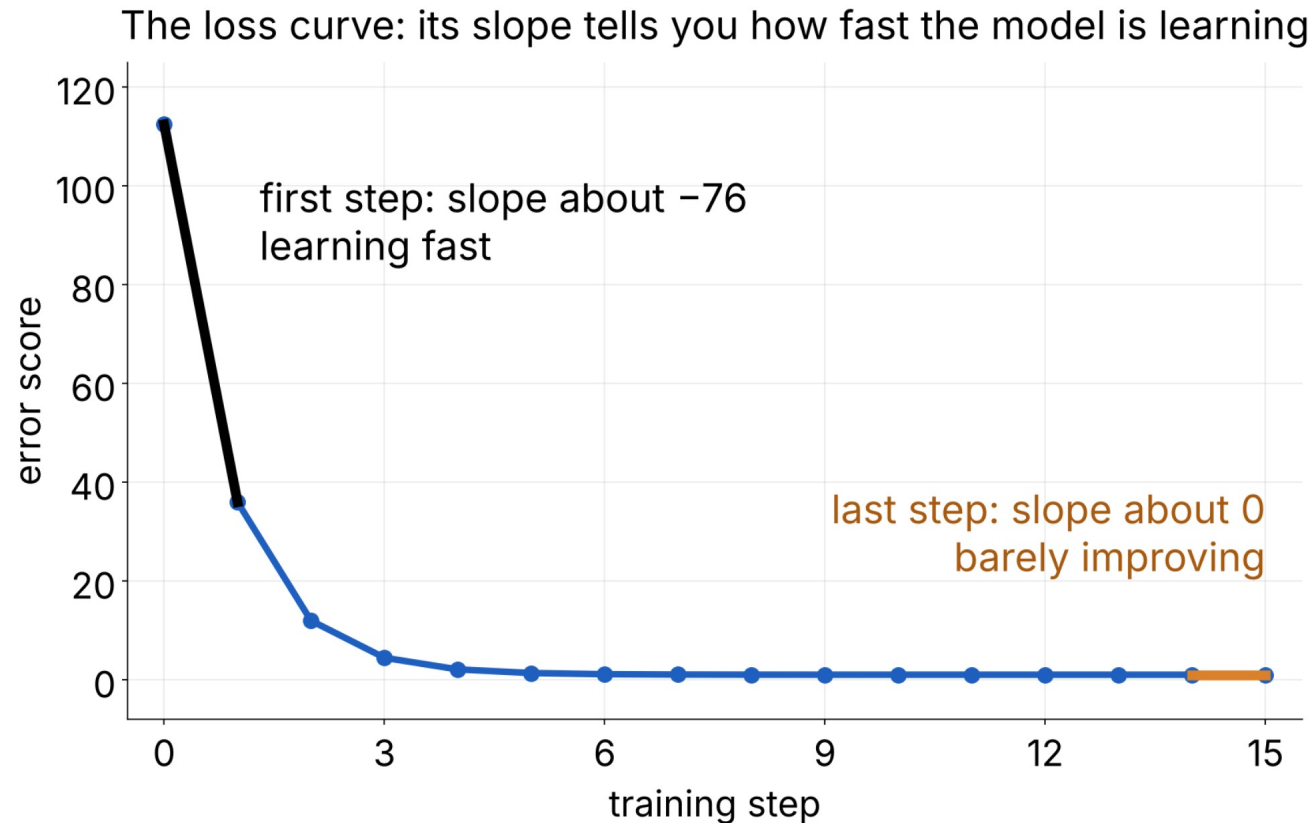
(Episode 0.1 called the valley slope “steepness.”)

The **minus** flips the sign, so you always step in the **downhill** direction.

Error: 112.4 → 35.96.

$$w_{\text{new}} = 5 - 0.02 \cdot (-70) = 5 + 1.4 = 6.4$$

Where slope lives in AI



SLOPE, TWO WAYS

To learn: every knob gets its own slope. Step each one downhill.

To watch: a loss curve's slope says how fast the model is still learning.

Your turn: find the slope

1 $(2, 1)$ and $(6, 13)$ What's the slope between them?

2 $y = -3x + 7$ What's its slope? Uphill or downhill?

3 $y = x^2$ from $x = 3$ to $x = 3.1$ Rise over run?

Pause the video. Say each one out loud, then work it out.

Answers

1 $\frac{13 - 1}{6 - 2} = \frac{12}{4} = 3$ Up 12, across 4: slope 3.

2 slope = $w = -3$ Negative, so downhill: 3 down per step right.

3 $\frac{3.1^2 - 3^2}{0.1} = \frac{9.61 - 9}{0.1} = 6.1$ Close to the slope at $x = 3$, which is 6.

THE EQUATION OF THE DAY

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope is how much the output changes for each step of the input. Its sign tells a model which way is downhill.

NEXT TIME • EPISODE 0.6

Exponents and Logarithms

Repeated multiplication, and how to undo it.

YOUR CHALLENGE • answer in Episode 0.6

$0.1 \rightarrow 0.2 \rightarrow 0.4 \rightarrow 0.8 \rightarrow \dots$ (mm)

Paper is 0.1 mm thick, and every fold doubles it. How thick after 10 folds? Bonus: how many folds to pass 1 meter?